Selection flags — the count doesn't matter

arxivkg · retrieval · ablation

What we did

`02-03` cleaned the pool with a cosine gate; this asks how to pick the final landmark set and **whether the cap must grow**. Four arms at cap 50, over all 27 topics, scoring *every* selected paper $-2..+2$ on its **salient full text** (Gemma-4-26B @T= 0): `sum` penalizes drift/noise (count-confounded), `mean` = per-paper quality (count-independent), `neg2_per_list` = outright-unrelated papers kept. Arms: `fixed50` (PPR top-50), `gate50` (0.35 cosine gate), `llm50` (drop role=unrelated over PPR-80, capped 50 — same count), `llm_dyn` (same filter, soft cap 75 — more count). Scores cached per `(topic, arxiv_id)`, config-independent.

Results (27 topics)

arm	count	sum	mean	unrel./list
fixed50	50.0	31.93	0.639	5.56
gate50	49.7	45.33	0.896	2.89
llm50	48.3	44.3	0.866	0.0
llm_dyn	70.5	65.74	0.876	0.0



Conclusion — the count doesn't matter; the filters are independent flags

Filtering raises both sum and mean: `fixed50`→`gate50` lifts sum 31.93 → 45.33 (+42%) and mean 0.639 → 0.896 (+40%); the unfiltered top-50 carries 5.6 unrelated papers per list, the gate halves that, the LLM filter zeroes it. **Going bigger raises sum via clean relevant mass while mean holds**: `llm50`→`llm_dyn` adds ∼22 papers, sum jumps 44.3 → 65.74 (+48%) while mean is flat (0.866 → 0.876) — the extra papers are genuine on-topic mass, not dilution, but they make no *paper* better. **So the count is not a lever**: `sum` rises with clean count (recall-flavored); `mean` — the count-independent quality — saturates at ∼0.88 once you filter, exactly where the compose-judge (`landmark-selection`, in `03-generation`) said the review saturates.

This makes cosine-gate and LLM-filter two independent on/off flags, not a tuned size: the 0.35 cosine gate alone gets ∼95% of the mean win for free (no LLM); the LLM filter drives residual unrelated papers from ∼3 to 0 (precision-max); either can be enabled without touching the cap. **Caveat:** `sum` is count-confounded by construction; `mean` is the fair cross-arm comparison. Downstream review quality is judged in `03-generation`.